BT 201 Mid Current Solved

MCQS

1. **Trypsin** ……..is used to cleave the protein.
2. Arabia thaliana's is…….. **Fly**
3. All peptide techniques are…….. **All**
4. **Mass spectrometry** …….. Is used to predict the protein structures.
5. Trochophore resembles like …….. **Larvae**
6. Plasmid size …….. **2 to 200 bp**
7. Humans differ from gorillas…….. **bipedal**
8. That's our transfer …….. **Traits**
9. Kimber Ella is …….. **Animal.**
10. Coral reef are disappeared due to …….. **Global Warming**
11. Vertebrates are …….. **5%**
12. **Micro ecology** …….. The possibility of working with molecules on the scale of a few micrometers,
13. Which institute was not involved in the human genome project? …….. **U.S...Center of health**

Question and Answers

How gene can be identified?

Disease **gene identification** is a process by which scientists **identify** the mutant genotypes responsible for an inherited **genetic** disorder. Mutations in these **genes can** include single nucleotide substitutions, single nucleotide additions/deletions, deletion of the entire **gene**, and other **genetic** abnormalities. **(Internet)**

In a fully sequenced genome, genes are found by scanning the sequence using gene-predicting computer programmes and assigning putative functions by searching for similarities in already existing databases. **(PPTS)**

Three point of phylogenetic? (Domain)

Bacteria, Archaea and Eukarya

What is bionomenclature?

In the 18th century, Carolus Linnaeus published a system of taxonomy based on resemblances. Two key features of his system remain useful today: two-part names for species and hierarchical classification. The two-part scientific name of a species is called a binomial. The first part of the name is the genus. The second part, called the specific epithet, is unique for each species within the genus. The first letter of the genus is capitalized, and the entire species name is italicized or underlined. Both parts together name the species. **(PPTS)**

Binomial nomenclature, also called binominal nomenclature or binary nomenclature, is a formal system of naming species of living things by giving each a name composed of two parts, both of which use Latin grammatical forms, although they can be based on words from other languages. **(Internet)**

Micro ecology?

Microecology means microbial ecology or ecology of a microhabitat. Human gut **microecology** is the study of microbial ecology of the human gut. **Microecology** is a large field which includes many topics such as evolution, biodiversity, exobiology, ecology, bioremediation, recycling, and food microbiology.

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Species diversity? Species Richness?

Species diversity is defined as the number of **species** and abundance of each **species** that live in a particular location. The number of **species** that live in a certain location is called **species** richness.

Different in ortholog and paralog gene?

Orthologs are **genes** that are related by vertical descent from a common ancestor and encode proteins with the same function in **different** species.

Paralogs are homologous **genes** that have evolved by duplication and code for protein with similar, but not identical functions."

Importance of bacteria and fungi?

Fungi and **bacteria** are essential to many of the most basic ecosystem processes: saprobic **fungi** break down fallen wood and litter returning nutrients to the soil; nitrogen-fixing **bacteria** and mycorrhizal **fungi** assist plants to obtain nutrients from the soil; and many groups of **fungi** and **bacteria** cause diseases of plants

History of plant breeding?

Gregor Mendel's experiments with plant hybridization led to his establishing laws of inheritance. Once this work became well known, it formed the basis of the new science of genetics, which stimulated research by many plant scientists dedicated to improving crop production through plant breeding

Bioinformatics?

Bioinformatics combines biology, computer science, information engineering, mathematics and statistics to analyze and interpret biological data. **Bioinformatics** has been used for in silico analyses of biological queries using mathematical and statistical techniques

Lack of fossil record?

At the time of Darwin, the fossil record was VERY scanty. A great deal of progress has been made since, but it is far from finished.

The fossil record is relatively incomplete for several reasons: 1. Soft tissues are rarely preserved 2. Movement of the earth's crust has obliterated and/or covered many fossils 3. Fossilization takes place only in certain types of habitats and favorable environments 4. Paleontologists have not dug up every place on earth

Enlist model?

Some models • Dynamic programming • Hidden Markov Models (HMMs) • Conditional random field (CRF) • Support vector machines (SVMs)

Evolution of organism documents from phylogeny? How evolutionary history convert into documents?

- Comparing nucleic acids or other molecules to infer relatedness is a valuable tool for tracing organisms' evolutionary history
- DNA that codes for rRNA changes relatively slowly and is useful for investigating branching points hundreds of millions of years ago
- mtDNA evolves rapidly and can be used to explore recent evolutionary events

Name phyla in amoeboid protozoa and their locomotion?

Protozoa can be divided into four **phyla** based on **their locomotion**: Mastigophora, Sarcodina, Ciliophora, and Sporozoa.

Systematics?

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systematics (cladistics) is a method of taxonomic classification based on their evolutionary history. It was developed by Willi Hennig, a German entomologist, in 1950.

Species richness Systematics?

Species richness is the number of different species in a particular community. If we found 30 species in one community, and 300 species in another, the second community would have much higher species richness than the first.

Phylogeny?

Phylogeny is the evolutionary history of a species or group of related species The discipline of systematics classifies organisms and determines their evolutionary relationships

Difference between arctic and polar region?

the **Arctic is** not a continent which **is** the predominant different **between** the two **polar regions**. Under the massive ice cap **of** the **Arctic** lies the **Arctic Ocean** whose depths reach 15,000 feet below the surface. Often times the two **polar regions** are confused due to their having such similar names. **(Internet)**

Law of succession?

General correspondence between fossil and living forms in same geographical area • Darwin used this observation to predict that fossils of ancient humans should be found in Africa

Bioinformatics systematics?

???

Why breeding crop important?

Crop breeding is the art and science of improving **important** agricultural plants for the benefit of humankind. **Crop breeders** work to make our food, fiber, forage, and industrial **crops** more productive and nutritious. **Crops** provide for an expanding global population with increasing dietary expectations. **(Internet)**

Factor affect commute structure?

The gender, age, number of family cars, housing source and jobs-housing balance are the most important **factors** influencing **commuting** mode choice; housing, population density, jobs-housing balance and **commuting** mode significantly **affect commuting** time; and transport accessibility, jobs-housing balance and **commuting** mode **(Internet)**

Ecology?

Ecology is a branch of biology that studies the interactions among organisms and their biophysical environment, which includes both biotic and abiotic components.

"Ecology is the scientific study of the processes regulating the distribution and abundance of organisms and the interactions among them, and the study of how these organisms in turn mediate the transport and transformation of energy and matter in the biosphere (i.e., the study of the design of ecosystem structure and function).

Applied Ecology?

Applied Ecology: Using ecological principles to maintain conditions necessary for the continuation of present day life on earth.

Plants breeding?

Plant breeding is the art and science of changing the traits of plants in order to produce desired characteristics. Plant breeding can be accomplished through many different techniques ranging from simply selecting plants with desirable characteristics for propagation, to methods that make use of knowledge of genetics and chromosomes, to more complex molecular techniques.

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Protozoa life cycle?

Protozoa have two-phase **life cycles**, alternating between proliferative stages (e.g., trophozoites) and dormant cysts. As cysts, **protozoa** can survive harsh conditions, such as exposure to extreme temperatures or harmful chemicals, or long periods without access to nutrients, water, or oxygen for periods of time. **(Internet)**

Five scientist citing reteach?

- **Kerner** showed that the differences in growth form observed by him and Bonnier were the result of temporary adjustments to environmental variation and not hereditary changes in the experimental plants.
- **Turesson's** studies provided evidence for genetic differences among populations.
- **Clausen, Keck, and Hiesey** explored the extent and sources of morphological variation in plant populations, including both the influences of environment and genetics.
- **Case** determined that the best predictor of chuckwalla, Sauromalus, body length was average winter rainfall **Tracy's** laboratory growth experiments indicated that variation in body size among chuckwalla populations is at least partly determined by genetic differences among populations.

How gene find sustainability of gene?

???

How climate affect community?

Climate change could **affect** our **society** through impacts on a number of different social, cultural, and natural resources. For example, **climate** change could **affect** human health, infrastructure, and transportation systems, as well as energy, food, and water supplies. **(Internet)**

Sequencing?

DNA **sequencing** is the process of determining the nucleic acid sequence – the order of nucleotides in DNA. It includes any method or technology that is used to determine the order of the four bases: adenine, guanine, cytosine, and thymine.

Long 5 Marks

Evolutionary related study to genome sequence?

Explore genome evolution based on large data sets of DNA or protein sequences. Using entire genomes to infer a species tree (Eisen and Fraser 2003). Based on maximum genetic information and average out the anomalies. Has become the standard for reconstructing reliable phylogenies (Ciccarelli et al, 2006; Daubin et al. 2002). **(Yehi Samjh aaya hai)**

Plant breeding?

Plant breeding is the science of changing the traits of **plants** in order to produce desired characteristics. It has been used to improve the quality of nutrition in products for humans and animals. ... **Plant breeding** has been practiced for thousands of years, since near the beginning of human civilization.

Why we use phylogenic tree? Why phylogenic tree is important? Reasons to use phylogenetic tree?

Phylogenetics is important because it enriches our understanding of how genes, genomes, species (and molecular sequences more generally) evolve.

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Biodiversity importance in term of food and human?

Biodiversity creates resilience and is a key to mitigate the risks in agriculture particularly farming, from where we get most of our **food**. It ensures the world gets nutrients which is critical for good **health** and is also critical for conserving the ecosystem that produces the **food**. **(Internet)**

Factor forming community shape?

Shape The Forces That Bring A Community Together

- Common goals. Common goals (whether long-term or short-term) unite members to contribute towards it. ...
- Common threat. Common threats, problems and enemies help forge groups together. ...
- Isolation from others. Isolation from others focuses the attention upon each other. ... Barriers to entry. Barriers to entry are important. **(Internet)**

Landscape ecology explain?

Landscape ecology is the science of studying and improving relationships between **ecological** processes in the environment and particular ecosystems. This is done within a variety of **landscape** scales, development spatial patterns, and organizational levels of research and policy. **(Internet)**

Linnaeus classification?

Carolus Linnaeus is the father of taxonomy, which is the system of classifying and naming organisms. One of his contributions was the development of a hierarchical system of classification of nature. Today, this system includes eight taxa: domain, kingdom, phylum, class, order, family, **genus**, and **species** **(Internet)**

Phylum Annelida?

Annelid, **phylum** name **Annelida**, also called segmented worm, any member of a **phylum** of invertebrate animals that are characterized by the possession of a body cavity (or coelom), movable bristles (or setae), and a body divided into segments by transverse rings, or annulations, from which they take their name. **(Internet)**

Phylum Annelida Class Polychaeta Family Siboglinidae Class Clitellata • Subclass Oligochaeta. Subclass Hirudinea

Annelid Characteristics • Defining Characteristics • One or more pairs of chitinous setae • The phylum includes polychaetes, earthworms, leeches, and vestimentiferans • True segmented worms • Metameric segmentation

Body Structure The body is a tube within a tube. The coelom is important to annelids for:The epidermis is what secretes the tough cuticle

Locomotion On each side of the animal is a parapod (parapodia) consisting of fleshly lobes, which are supported by chitinous rods. Each parapod have setae, which can be sharp (protection), and aid in locomotion

Feeding Annelids range from carnivores, herbivores, scavengers, deposit feeders, and filter feeders. With very few defenses, many remain in a burrow or secreted tube. Carnivores can capture prey with strong jaws and quickly drag it back to its burrow. Can use a muscular pharynx = eversible proboscis. **(PPTS)**